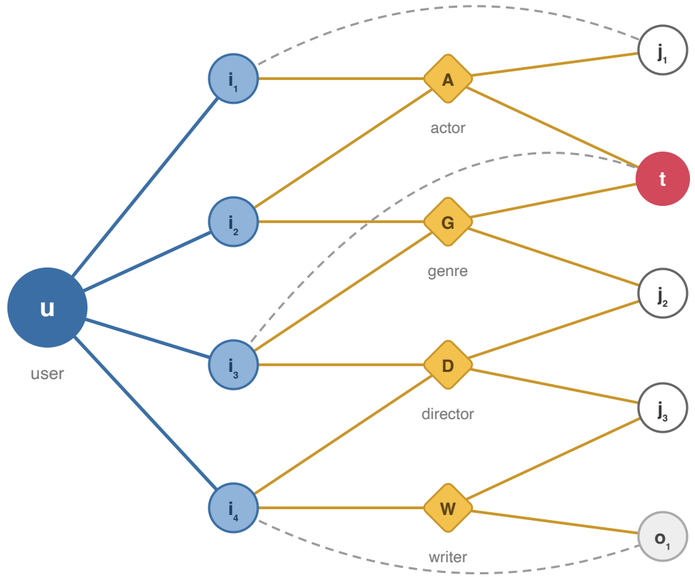
a Original recommendation



score & rank

Top-K list

rank	
1	j_1
2	t
3	j_2
4	j_3
⋮	
10	j_9
K = 10	
	o_1 outsider

KG path · KG entity · collaborative pa · target t

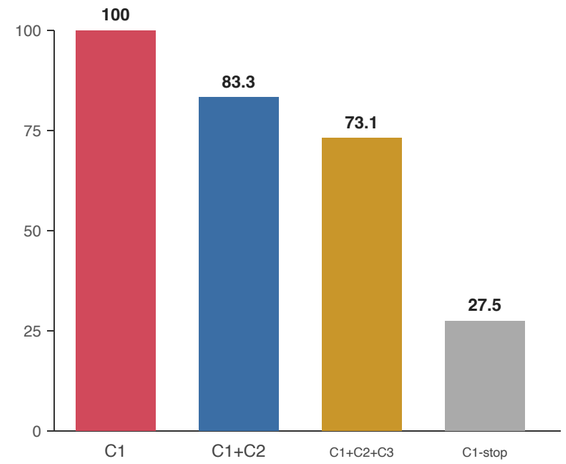
Request: remove t, keep the rest

- C1 t leaves top-K
- C2 other K–1 items stay
- C3 each keeps ≥1 evidence path

min |M| s.t. C1+C2+C3

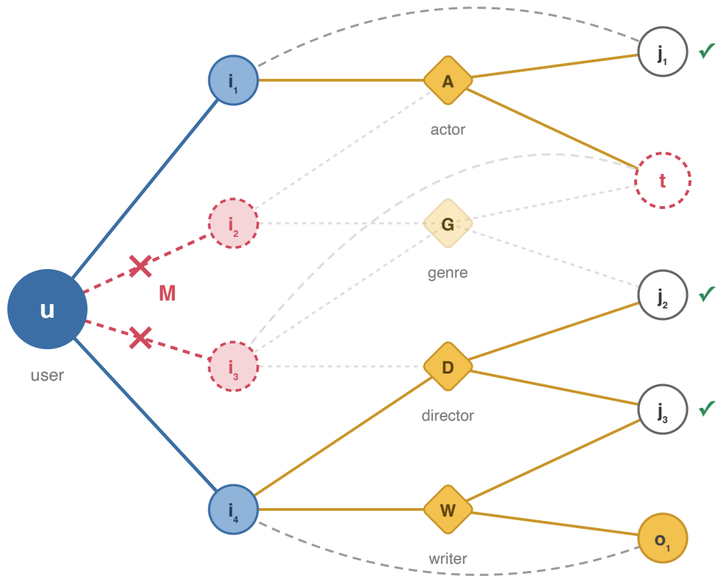
exact MILP · lazy constraints
every mask verified by re-ranking

MovieLens-1M: valid mask (%)



apply M

b Counterfactual with mask M = {i_2, i_3}



re-rank

Counterfactual top-K

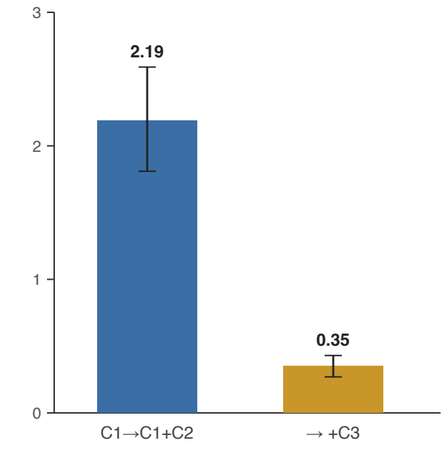
rank		
1	j_1	✓
2	j_2	✓
3	j_3	✓
⋮		
9	j_9	✓
10	o_1	new
K = 10		
	t	removed

Outcome

- C1 t removed ✓
- C2 j_1 … j_9 remain; o_1 enters ✓
- C3 j_1 via A · j_2 via D · j_3 via W ✓

Explanation: "without i_2 and i_3, t would not be recommended"

Extra interactions masked (ΔIMI)



paired means, 95% bootstrap CI